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Industrial fans: improving system reliability

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Introduction



Source: Europol, Flakt

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Objectives

By the end of this module, you will have a better understanding of:

- Fan and blower types
- Bearings and housing typically used in fans
- Self-aligning bearing system for fans
- Proper installation of bearings, including recommended ISO shaft and housing fits
- Lubrication of fan bearings
- Diagnosing common fan problems

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Overview

Background information concerning fans and blowers including:

- Bearing selection
- Bearing arrangements
- Fan types
- Fan function
- Fan terminology
- Lubrication
- Common fan problems
- Condition monitoring practices
- Mounting/dismounting

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Fan basics

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What is a fan?

- Any device consisting of a series of radiating vanes or blades attached to and revolving with a central hub-like portion and used to produce a current of air. *



* American College Dictionary.

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What are fans used for?

Some common uses:

- Heating, cooling, ventilation
- Drying
- Moving materials
- Humidify, dehumidify
- Induce draft
- Filtration
- Atomize and propel liquids



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Fans are everywhere!

Industrial:

- Mine shaft ventilation
- Pulp mill chip blowers
- Tissue mill through air dryers
- Fluid bed dryers
- Food industry dryers



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How a fan works

A fan is a device that creates flow by using a drive system to rotate blades

- Axial flow fans use a propeller to create flow in the direction of the axis of rotation
- Centrifugal blowers move gas by means of the centrifugal force generated by rotating a cylindrical runner with blades
- Cross flow fans create a wide flow of gas using a long cylindrical-shaped impeller. Gas is drawn in along the sides of the runner, flows along the circumference of the runner, and is blown out laterally

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Fan types



Centrifugal Blowers



Axial Flow Fans



Cross Flow Fans

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Axial flow fans (propeller type)

- The axial flow fan increases air velocity through a rotating force that produces velocity pressure or kinetic energy with only a small increase in potential energy



Pedestal Type

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Axial fans (belt & direct drive)



Belt



Direct

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Centrifugal fans and blowers

- Centrifugal fans have a scroll-shaped housing
- Gas is pulled in axially and flung out radially
- Direct or belt driven



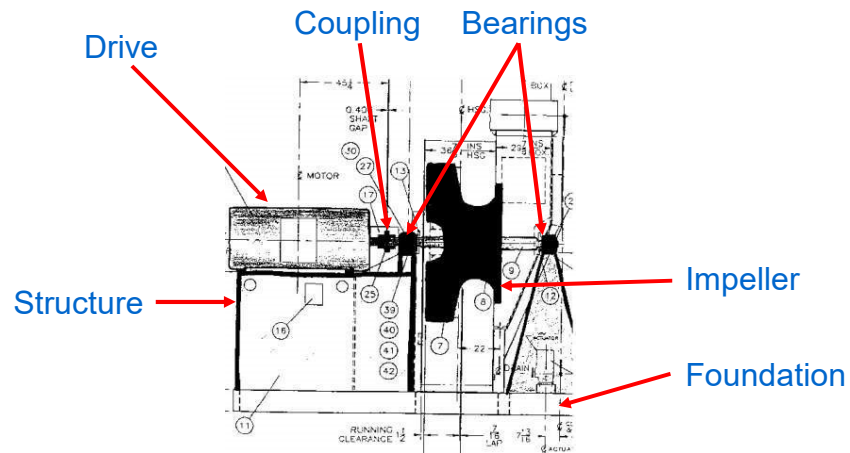
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Centrifugal flow fan components



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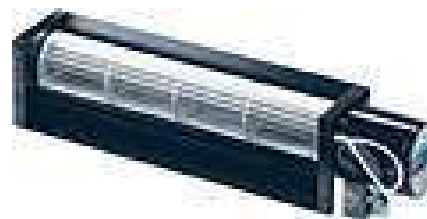
Cross flow fans

Applications:

- Portable room heaters
- Circuit board cooling
- Cooling columns

Advantages:

- Wide sheet of air
- Uniform flow
- Low profile – space saving



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Centrifugal fans (belt and direct drive)



Belt



Direct

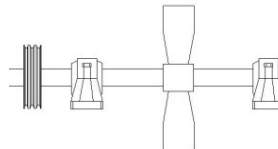
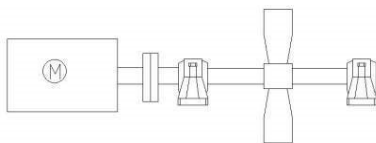
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Direct versus belt drive

- Direct drive couplings can apply load in any direction depending on alignment
- Coupling can add unbalance
- Coupling can apply a thrust load
- Belt drives add a highly variable radial load that can often yield a lightly loaded bearing opposite the belt side
- Belts and their pulleys can also be out of balance



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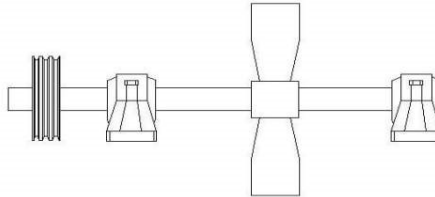
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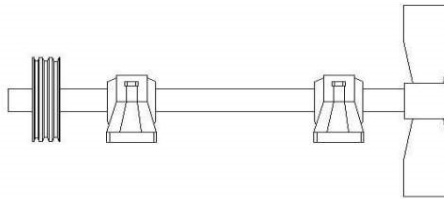
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Overhung versus straddle mount...

Center-hung,
Belt Driven



Overhung,
Belt Driven
(Most Common)



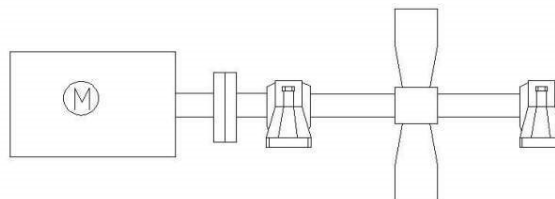
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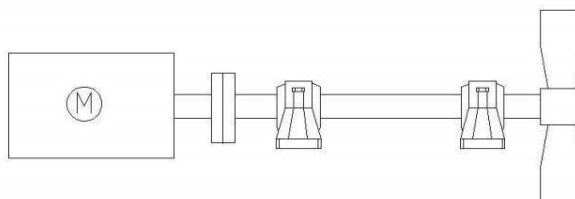
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Overhung versus straddle mount...

Center-hung
(straddle-mount),
Direct Drive



Overhung,
Direct Drive



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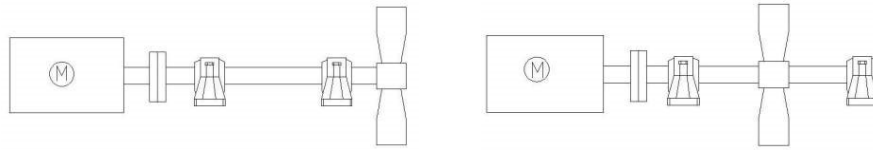
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Overhung versus straddle mount

- Overhung fans - impeller beyond both bearings on the shaft
- Overhung fans can have very lightly loaded bearings
- Sensitive to unbalance, fretting, minimum load
- Center hung fans – impeller between the bearings
- Center hung fans often have large thermal expansion of the shaft (hot air/gas)



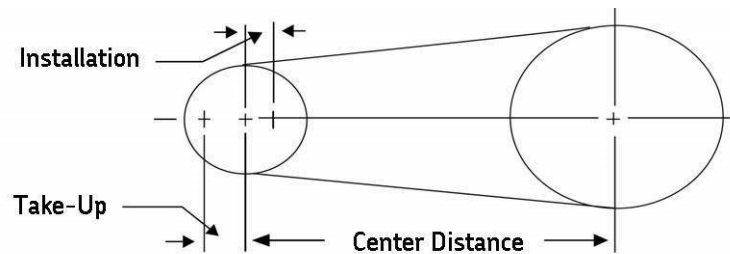
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Belt troubleshooting

- Verify that there is sufficient allowable movement or if drive was at an extreme of movement. Belt manuals contain charts for minimum take-up and installation distances based on belt length



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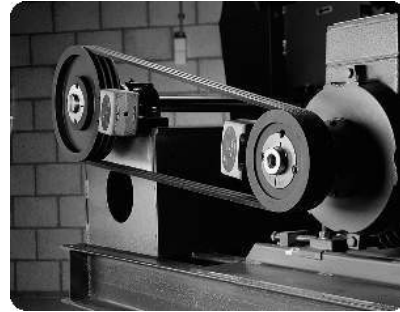
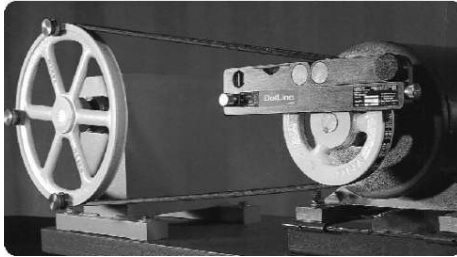
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Best practices: fan sheave laser alignment

Use of laser alignment devices



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Fan bearing guidelines

- “C3” radial internal clearance recommended
- In general, the ratio of axial to radial should not be:
 - > 1 for spherical roller bearings
 - > 0.25 for self-aligning ball bearings
- Check minimum radial load
- Labyrinth seals may be required due to high speeds

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Bearing life

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Bearing life: selecting bearing size

- The catalog value (C) predicts a 10% chance of failure at one million revolutions of operation
 - If the bearings were operated at the catalog rated load for one million revolutions, 90% of the bearings would still be running
- This is also known as “L10” Life
- Average Bearing Life (L50) is five times L10

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Predicting bearing life

- Designer reviews the application and designs to desired life
 - Life for a vacuum cleaner might be 1,000 hours
 - Life for typical industrial applications is about 40,000 hours, or 4-5 years
- Costlier machines typically designed for longer lives
 - Paper machine bearings are designed for 100-200,000 hours life (10-20 years)

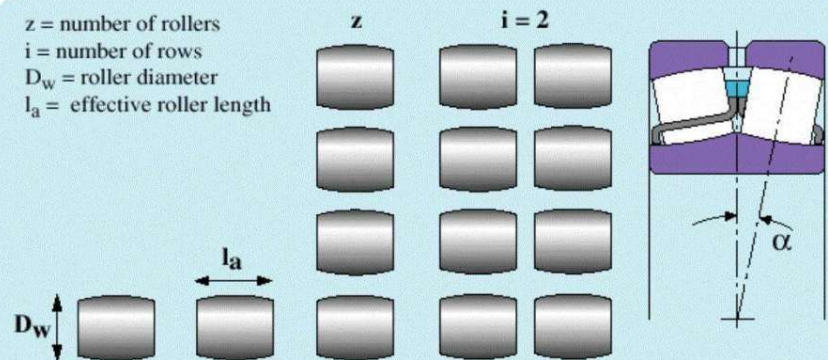
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ISO/ABMA life rating

z = number of rollers
 i = number of rows
 D_w = roller diameter
 l_a = effective roller length



$$C = \text{func}(D_w, l_a, z, i, \alpha)$$

C is based on macro-geometry only!

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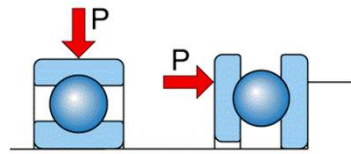
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Bearing load capacity

Basic dynamic load rating C



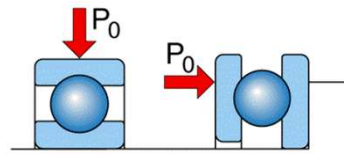
The ISO life equation

$$L_{10} = \left(\frac{C}{P} \right)^p$$

L_{10} = basic rating life, millions of revolutions
 C = basic dynamic load rating, N
 P = equivalent dynamic bearing load, N
 p = exponent of the life equation

With the load $P = C$ the L_{10} life will be 1 million revolutions

Basic static load rating C_0



The static safety factor

$$s_0 = \frac{C_0}{P_0}$$

s_0 = static safety factor
 P_0 = equivalent static bearing load, N
 C_0 = basic static load rating, N

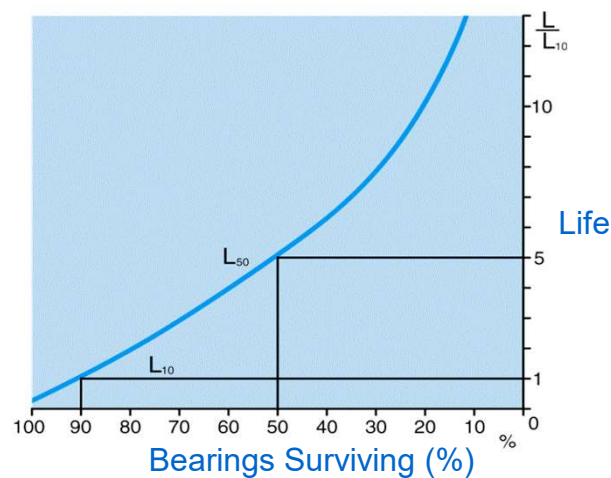
With the load $P_0 = C_0$ the static safety factor s_0 will be 1

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Bearing life...



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Bearing life

- What is **one million revolutions**?
- Consider an old-style record player at 33-1/3 RPM
 - At 33-1/3 RPM, how many hours do I have to run before getting one million revolutions?
- One million revolutions divided by 33-1/3 revolutions per minute = 30,000 minutes = 500 hours
- This may be acceptable life: ten hours or less per week for 50 weeks would be 500 hours
 - Meets a one year warranty requirement

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Basic bearing life calculation

$$L_{10} = (C/P)^p$$

- $p = 3$ for ball bearings, $10/3$ for roller bearings
- C = Basic Dynamic Load rating (from the bearing catalog)
- P = Equivalent Dynamic Load applied to the bearing

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Bearing life calculation...

$$L_{10h} = \frac{1,000,000 \times (C/P)^p}{(60 \times n)}$$

- L_{10h} = Basic life in hours
- 60 – 60 minutes/hour
- n = Revolutions per minute (RPM)

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Bearing life calculation...

Example problem:

- A deep groove ball bearing with a L_{10h} life of 50,000 hours is required for an operating speed of 3,600 rpm (n) and radial load of only 1,000lbs (P)
- What bearing capacity is chosen?

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Bearing life calculation

- Solution: Solve the life equation for “C”
- Use the equation and a calculator

$$L_{10h} = \frac{1,000,000 \times (C/P)^p}{(60 \times n)}$$

$$C = 22,104 \text{ KN}$$

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Fan bearing arrangements

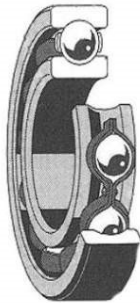
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Fan bearings (ball bearings)



Single Row Deep
Groove Ball
Bearing (DGBB)



Single Row
Angular Contact
Ball Bearing
(SRACBB)



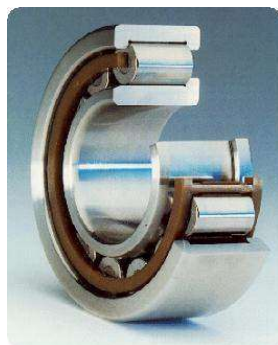
Self-Aligning
Ball Bearing
(SABB)

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Fan bearings (roller bearings)



Cylindrical roller



Spherical roller



Compact aligning
roller bearing (CARB)

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Fan bearing housings



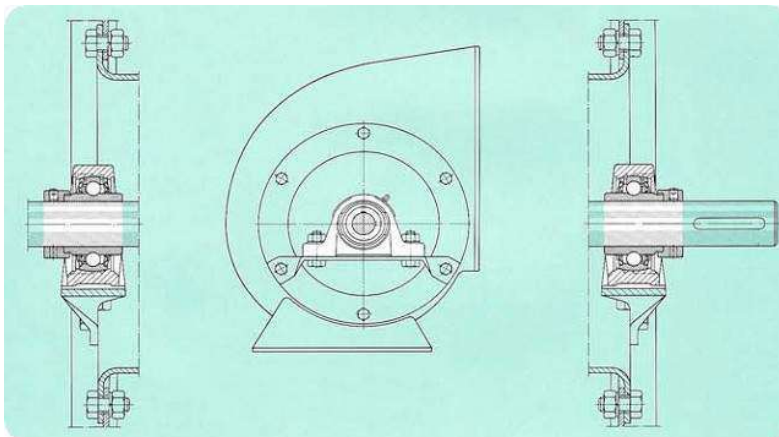
Unit ball bearing pillow blocks and flanges

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Unit ball bearing pillow blocks and flanges



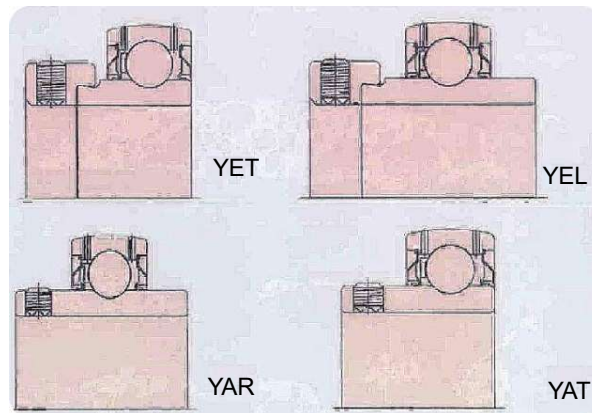
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Ball bearing inserts



Unit block bearing insert types

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Unit bearing housings with roller bearings



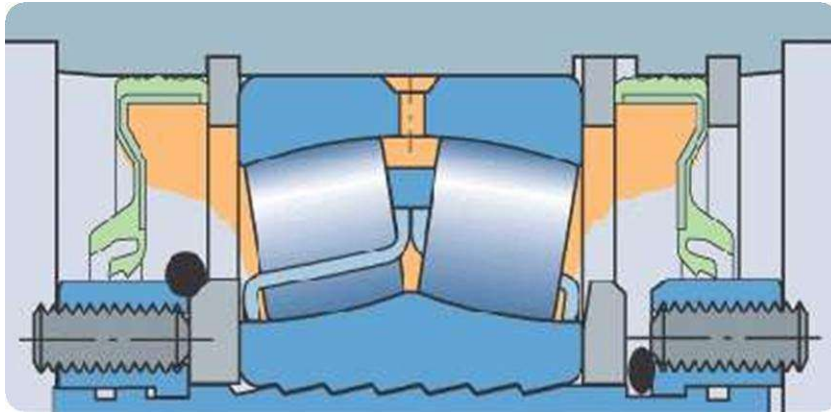
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ConCentra: cross section

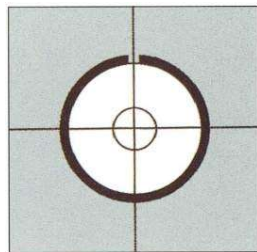


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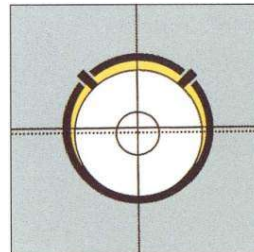
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ConCentra versus eccentric locking collar



ConCentra
Mounting



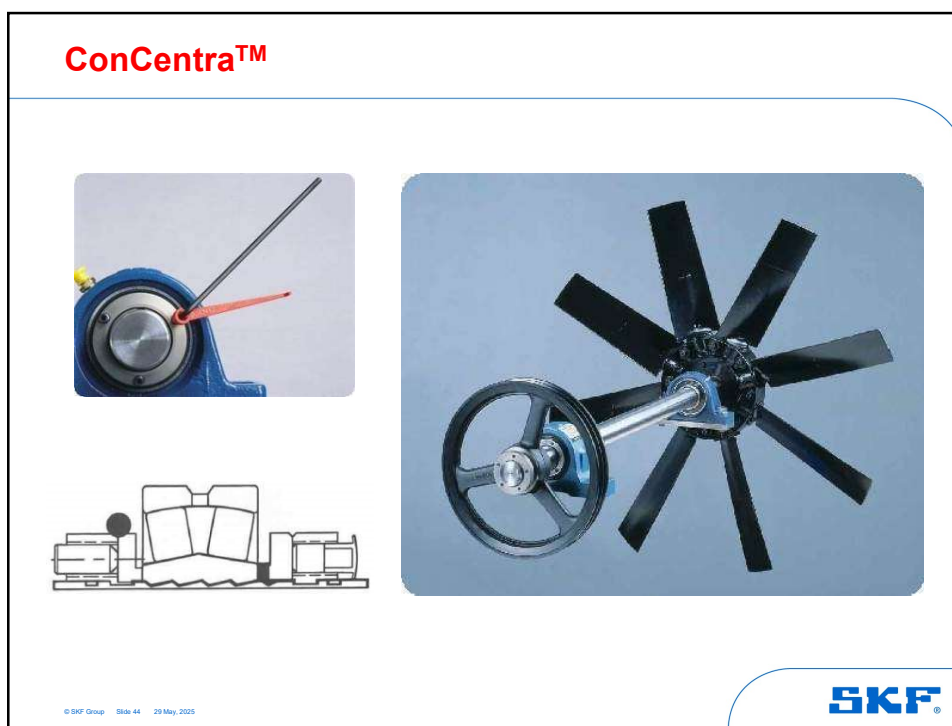
Eccentric Locking Collar

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Split bearing housings

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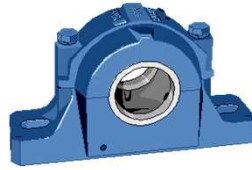
Fan bearing heavy-duty housings...

SNL



Metric Dimensions

SAF



Inch Dimensions

- Both types can accept either straight or stepped shafts
- Both types accept metric or imperial shafts
- Both can accept SABB, SRB, or CARB
- Base-to-center, bolt hole spacing, overall dimensions are not interchangeable

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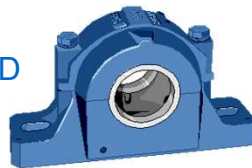
Fan bearing heavy-duty housings

SNL



Metric Dimensions

SAFD



Inch Dimensions

- Both types can accept either straight or stepped shafts
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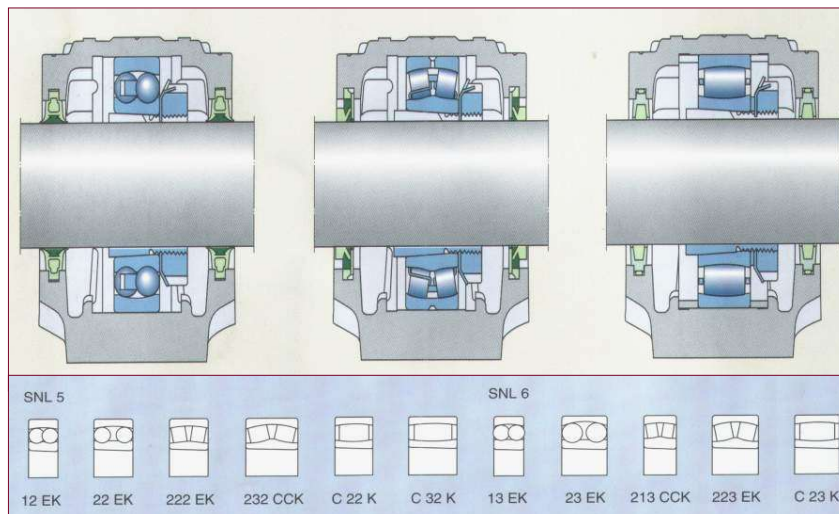
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What bearings fit in split housings?...



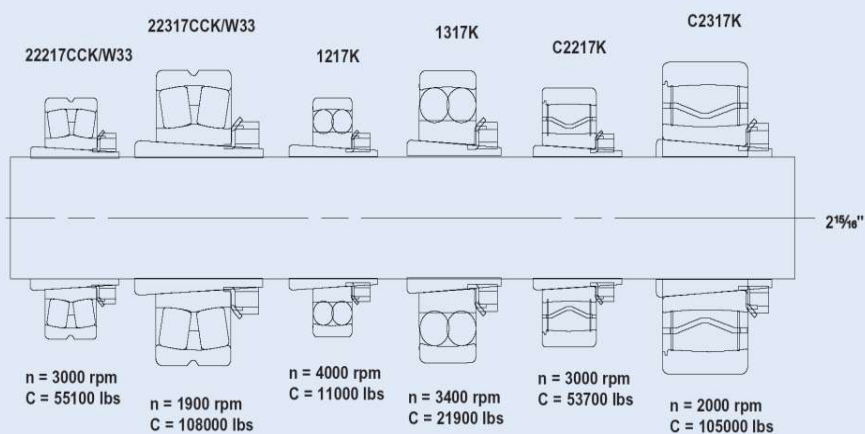
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What bearings fit in split housings?

Multiple bearing types and series accommodate a variety of applications.



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SNL metric pillow blocks

- Accept SRB, SABB, CARB
- Two styles:
 - Tapered bore
 - Cylindrical bore
- External dimensions are metric
- Metric or imperial dimension shafts are accommodated

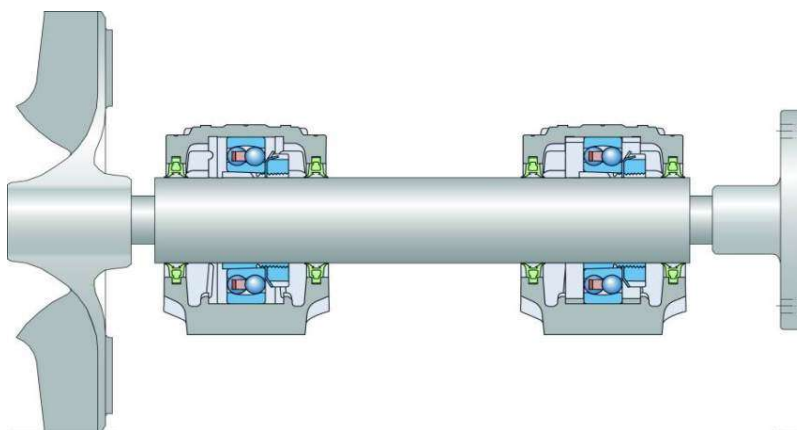


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SNL housings with SABB



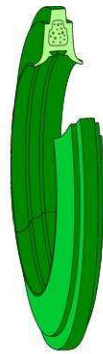
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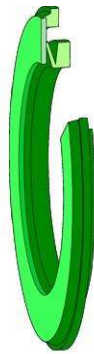
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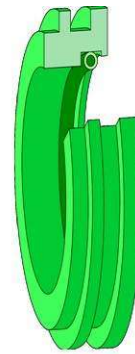
SNL metric pillow block seals



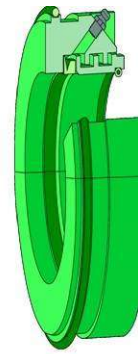
Double-lip



V-ring



Labyrinth



Taconite

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TSN .. L compared to TSN .. G

	 TSN .. L	 TSN .. G
Material	Injection molded TPE	Cast polyurethane
Temperature	-40° C to +100° C	-40° C to +100° C
Peripheral Speed	13 m/s	8 m/s (re-lubrication 4 m/s)
Misalignment	0.5 – 1°	0.5 – 1°

Sealing efficiency equal to TSN .. G

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SONL: for oil lubrication



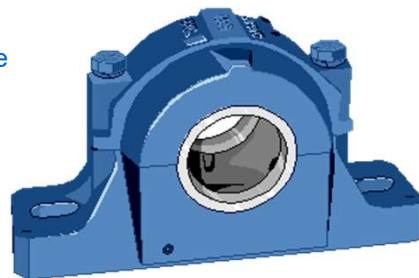
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SAF and SAFD inch dimensioned split pillow blocks

- These housings accept:
 - SRB, SABB, CARB bearings
- Two styles, two widths:
 - Tapered bore or Cylindrical bore
 - Narrow (5xx) or Wide (6xx) Series
 - External dimensions are inches
 - Metric or imperial dimension shafts



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Standard seals for SAF and SAFD



LOR (Current Style)

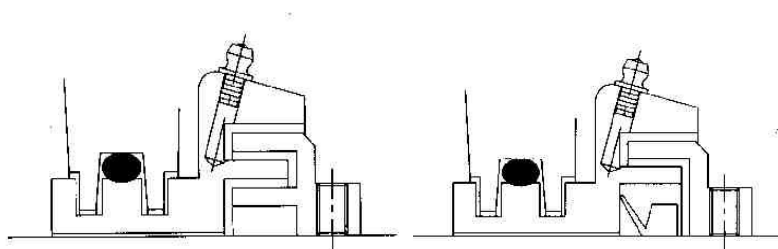
LER
(Replaced)

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SAFD extreme service seals



TER-C

TER-CV

Labyrinth extreme service seals – Taconite Seal

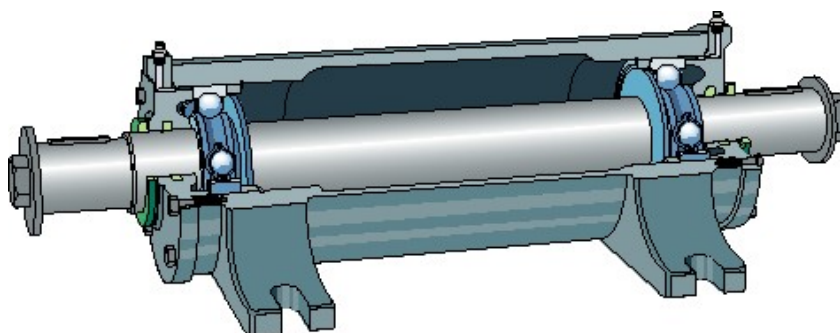
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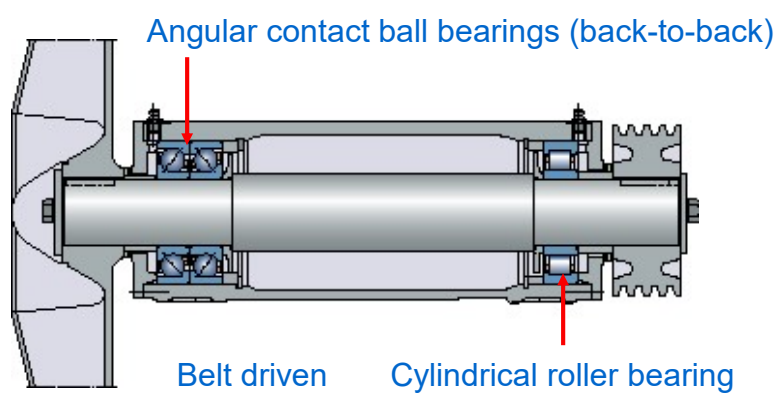
Two bearing housings: PDN unit



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Two bearing housings...

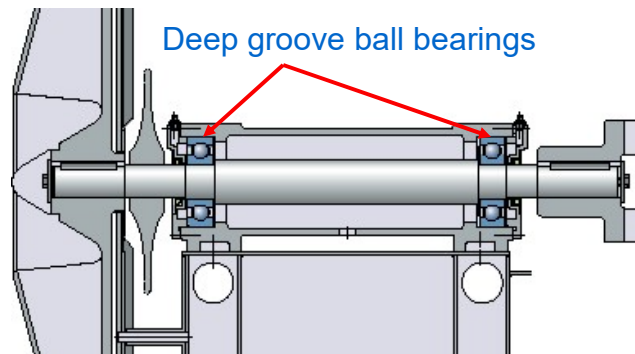


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Two bearing housings



Direct coupled

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Dismounting

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Dismounting objectives

- Prevent damage to the shaft or housing
- Save the bearing for remounting or rework
- Save parts for failure analysis

Rework: generally for bearings > 250mm O.D

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Safety...

- Fluoroelastomers are especially hazardous
 - Excess heat (~300° C, 600° F) can produce hydrofluoric acid fumes
 - Melted material will burn through clothing and skin

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Safety

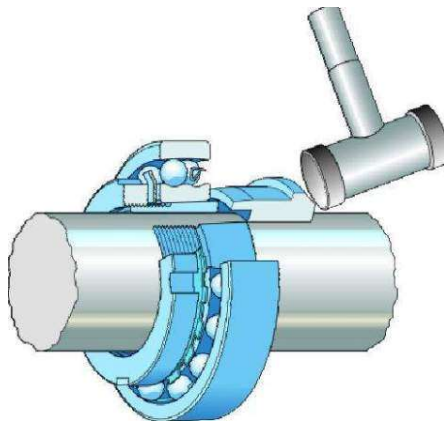
- Minimum protective gear for removal using non-recommended methods
 - Full face shield and safety glasses
 - Full coveralls and gloves
- Seal precautions
 - Remove the seal before torching by cutting through the seal and peeling out of the bearing

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Dismounting from a tapered journal or adapter sleeve



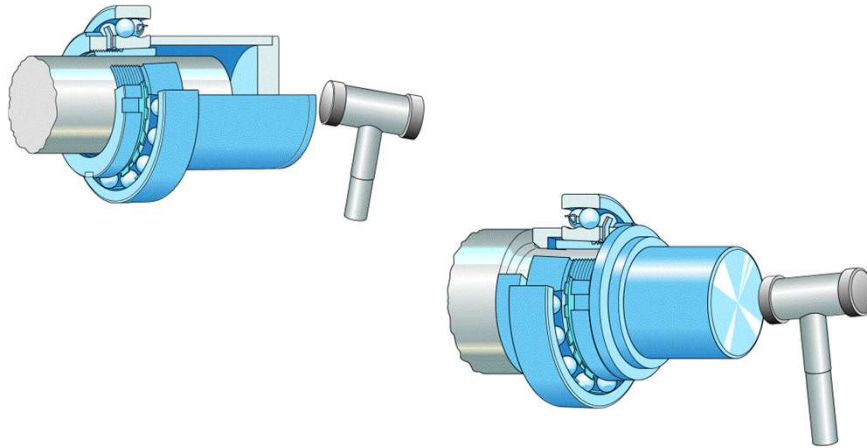
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Dismounting force



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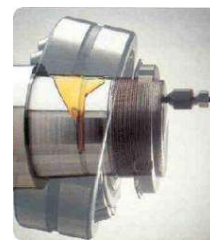
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Dismounting methods

Hydraulic Assist

- Advantages:
 - Quick and easy
 - Only requires a hydraulic pump
 - No damage to the bearing
 - No damage to the shaft
- Disadvantages:
 - Rare, not used by OEM's much
 - Added cost of preparing shaft with groove and ducts



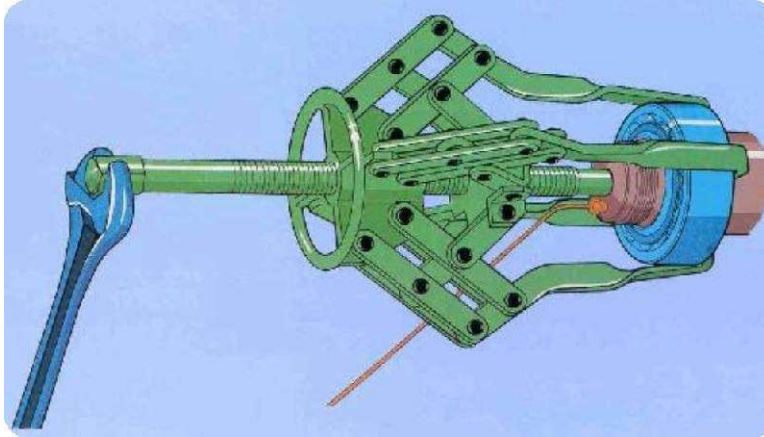
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Hydraulic cylindrical bore dismounting

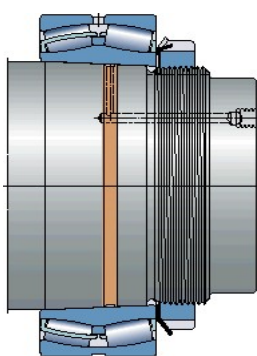


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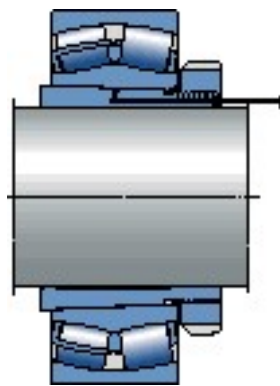
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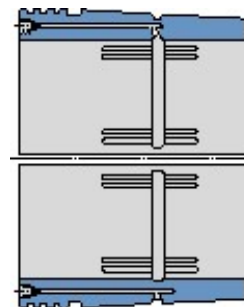
Hydraulic tapered dismounting



Tapered Shaft



Tapered Adapter
Sleeve



Tapered Withdrawal
Sleeve

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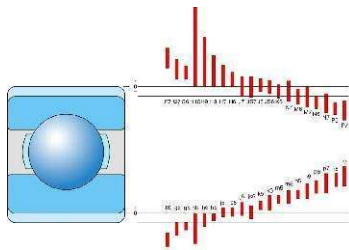
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Fits used for fan bearings: unit blocks

- Shaft tolerance generally **h9/IT5** for mounting on drawn steel or cold rolled shafting
- Eccentric operation leads to vibration and fretting corrosion of the bearing bore
- Suitable for low to medium speeds



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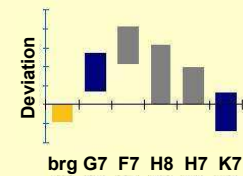
70

Fits for SNL housings: adapter sleeve, tapered bore bearings

- Shaft tolerances are generally **h9/IT5**
- Housing fits for clearance fit – **G7**
- For intermediate/tight fit – **K7**
- Benefits:
 - Elimination of housing wear and fretting corrosion
 - Improved reliability



Housing Bore Tolerances



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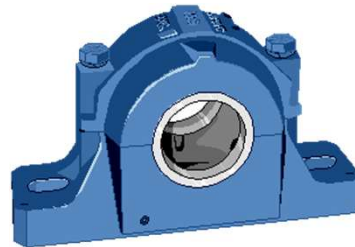
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Fits for SAF, SAFD housings: adapter sleeve, tapered bore bearings

- Shaft tolerances are generally h9/IT5
- Housing fits for clearance fit – G7
- SKF USA builds to G6 Standard for many sizes
- Some Clients of SKF: H7
 - Reports of reduced vibration



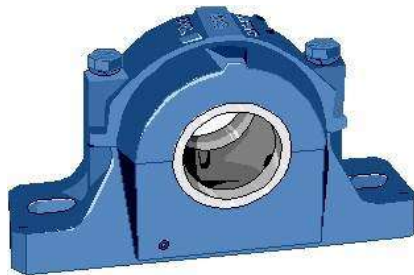
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Fits for SNL and SAFD: step shafts, parallel bore bearings

- Select fits for application, type, and size of bearings from ISO tables



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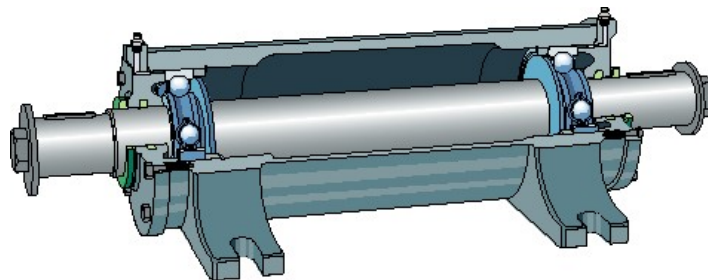
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Fits for two bearing housings: PDN, PDR, PDPF

Shaft fit – k6

Housing fit – G6



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Mounting and dismounting

- Pullers
- Fitting tools
- Heaters



- Oil power
- Instruments
- Lubricants



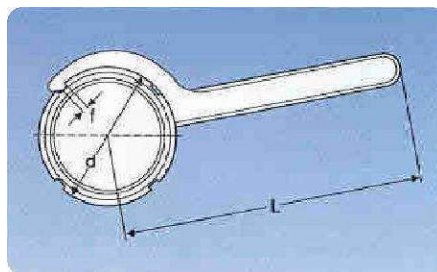
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Hook spanners

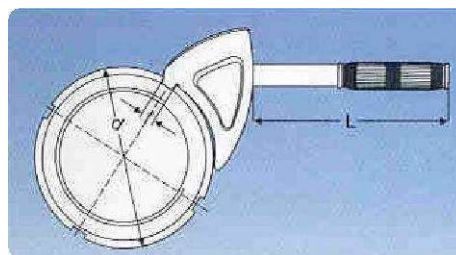


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Impact spanners



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Accessories

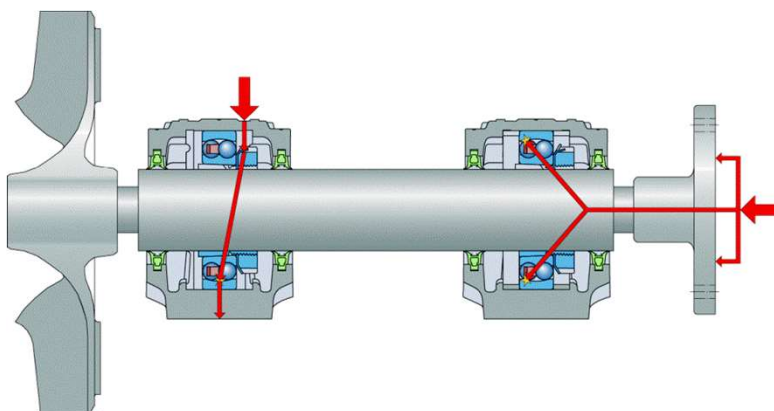


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Fitting on-site: take care



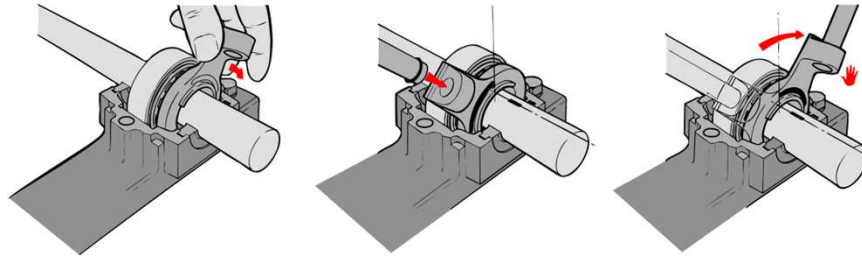
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Sleeve mounting of self-aligning ball bearings



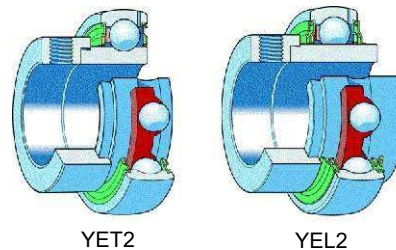
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Mounted unit installation

- Use precision ground shafting in heavier loaded applications
- Anti-seize compounds promote shaft creep and are not recommended
- Two mounting methods:
 1. Eccentric collar mounting
 - Set the collar in direction of shaft rotation
 - Apply set-screw(s) after locking in place
 2. Set-screw mounting
 - Torque to recommended values



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Angle drive-up method...

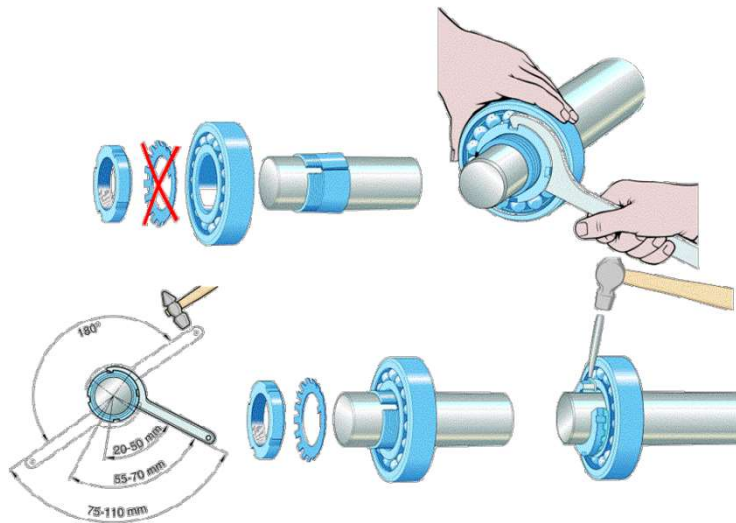
- Place the bearing on the shaft
- Apply the nut
- Drive the bearing up the tapered sleeve or shaft journal
- Remove the nut
- Secure the bearing with locknut and lock-washer

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Angle drive-up method



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Mounting spherical roller bearings with tapered bore



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Clearance reduction method

- Measure and record bench clearance using feeler gauges
- Determine the clearance reduction
- Drive-up the bearing with the nut until the reduction is achieved
- Remove the nut
- Secure the bearing with locknut and lock-washer



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Clearance reduction method: feeler gauges

Example: 22328 CCK/C3/W33

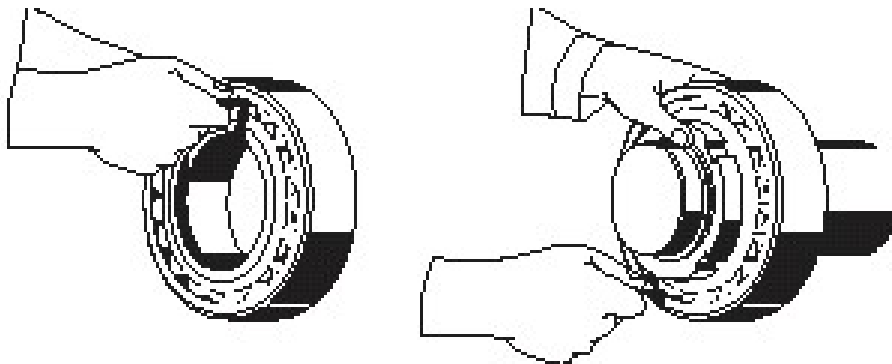
- K = Tapered bore
- C3 = Clearance of Bearing
- 28 x 5 = Bore = 140 mm

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Clearance reduction method...



Class exercise:

- Determine clearance reduction values for several spherical roller bearings

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Clearance reduction method

Class exercise:

- Refer to SKF Installation and Maintenance Guide, 140-710 Page 10 or plastic clearance reduction cards

order to eliminate creeping in this case, the outer ring must be mounted with a press fit in the housing. The rotating inner ring, when subjected to a rotating load, can be mounted with a slip fit on the shaft.

When the ring rotates in relation to the load, a tight fit is required.

Tapered Bore Bearings

The mounting of any tapered bore bearing is affected by giving the bearing on its seat a suitable amount

This amount may be determined either by measuring the reduction of internal radial clearance or the axial draw-up.

Draw-up is achieved with a force of sufficient magnitude applied directly to the face of the ring with the interference fit. This force is generated with one of the following devices:

1. Threaded lock nut
2. Ball end plate
3. Hydraulic nut
4. Mounting sleeve



How to measure unmounted radial internal clearance of SKF tapered bore spherical roller bearings (in inches)

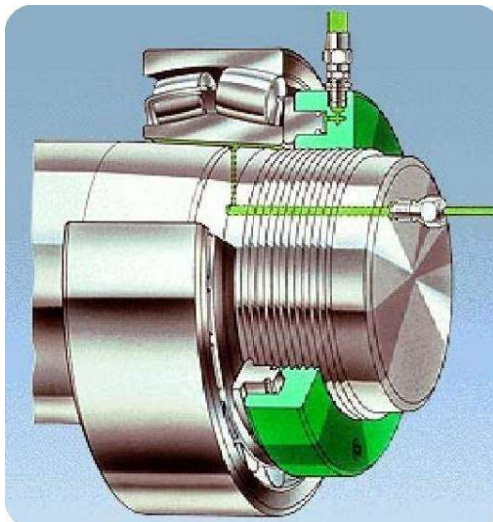
Outline the inner ring in a circumferential direction to properly seat the rollers. Measure the radial internal clearance in the bearing by inserting progressively larger feeler blades the full length of the roller between the most polished roller and the outer ring splines. CAUTION: Do not oil-lead the feeler blade, slide it through the clearance. Record the measurement on the largest size blade that will slide through. This is the unmounted radial internal clearance.

Repeat this procedure in two or three other locations by rotating the bearing on a different spot on its O.D. and measuring over different rollers in one row. Repeat the above procedure for the other row of rollers or measure each row alternately in the procedure described above (See chart A).

Chart A — Unmounted Radial Internal Clearance of SKF Tapered Bore Spherical Roller Bearings (in inches)

Bore Dia.		Normal		C3		C4	
Over	Ind.	Min.	Max.	Min.	Max.	Min.	Max.
34	30	0.0012	0.0010	0.0008	0.0002	0.0002	0.0000
40	35	0.0014	0.0012	0.0009	0.0002	0.0002	0.0000
46	41	0.0016	0.0013	0.0010	0.0002	0.0002	0.0000
52	47	0.0018	0.0014	0.0011	0.0002	0.0002	0.0000
58	53	0.0020	0.0015	0.0012	0.0002	0.0002	0.0000
64	59	0.0022	0.0016	0.0013	0.0002	0.0002	0.0000
70	65	0.0024	0.0017	0.0014	0.0002	0.0002	0.0000
76	71	0.0026	0.0018	0.0015	0.0002	0.0002	0.0000
82	77	0.0028	0.0019	0.0016	0.0002	0.0002	0.0000
88	83	0.0030	0.0020	0.0017	0.0002	0.0002	0.0000
94	89	0.0032	0.0021	0.0018	0.0002	0.0002	0.0000
100	95	0.0034	0.0022	0.0019	0.0002	0.0002	0.0000
106	101	0.0036	0.0023	0.0020	0.0002	0.0002	0.0000
112	107	0.0038	0.0024	0.0021	0.0002	0.0002	0.0000
118	113	0.0040	0.0025	0.0022	0.0002	0.0002	0.0000
124	119	0.0042	0.0026	0.0023	0.0002	0.0002	0.0000
130	125	0.0044	0.0027	0.0024	0.0002	0.0002	0.0000
136	131	0.0046	0.0028	0.0025	0.0002	0.0002	0.0000
142	137	0.0048	0.0029	0.0026	0.0002	0.0002	0.0000
148	143	0.0050	0.0030	0.0027	0.0002	0.0002	0.0000
154	149	0.0052	0.0031	0.0028	0.0002	0.0002	0.0000
160	155	0.0054	0.0032	0.0029	0.0002	0.0002	0.0000
166	161	0.0056	0.0033	0.0030	0.0002	0.0002	0.0000
172	167	0.0058	0.0034	0.0031	0.0002	0.0002	0.0000
178	173	0.0060	0.0035	0.0032	0.0002	0.0002	0.0000
184	179	0.0062	0.0036	0.0033	0.0002	0.0002	0.0000
190	185	0.0064	0.0037	0.0034	0.0002	0.0002	0.0000
196	191	0.0066	0.0038	0.0035	0.0002	0.0002	0.0000
202	197	0.0068	0.0039	0.0036	0.0002	0.0002	0.0000
208	203	0.0070	0.0040	0.0037	0.0002	0.0002	0.0000
214	209	0.0072	0.0041	0.0038	0.0002	0.0002	0.0000
220	215	0.0074	0.0042	0.0039	0.0002	0.0002	0.0000
226	221	0.0076	0.0043	0.0040	0.0002	0.0002	0.0000
232	227	0.0078	0.0044	0.0041	0.0002	0.0002	0.0000
238	233	0.0080	0.0045	0.0042	0.0002	0.0002	0.0000
244	239	0.0082	0.0046	0.0043	0.0002	0.0002	0.0000
250	245	0.0084	0.0047	0.0044	0.0002	0.0002	0.0000
256	251	0.0086	0.0048	0.0045	0.0002	0.0002	0.0000
262	257	0.0088	0.0049	0.0046	0.0002	0.0002	0.0000
268	263	0.0090	0.0050	0.0047	0.0002	0.0002	0.0000
274	269	0.0092	0.0051	0.0048	0.0002	0.0002	0.0000
280	275	0.0094	0.0052	0.0049	0.0002	0.0002	0.0000
286	281	0.0096	0.0053	0.0050	0.0002	0.0002	0.0000
292	287	0.0098	0.0054	0.0051	0.0002	0.0002	0.0000
298	293	0.0100	0.0055	0.0052	0.0002	0.0002	0.0000
304	299	0.0102	0.0056	0.0053	0.0002	0.0002	0.0000
310	305	0.0104	0.0057	0.0054	0.0002	0.0002	0.0000
316	311	0.0106	0.0058	0.0055	0.0002	0.0002	0.0000
322	317	0.0108	0.0059	0.0056	0.0002	0.0002	0.0000
328	323	0.0110	0.0060	0.0057	0.0002	0.0002	0.0000
334	329	0.0112	0.0061	0.0058	0.0002	0.0002	0.0000
340	335	0.0114	0.0062	0.0059	0.0002	0.0002	0.0000
346	341	0.0116	0.0063	0.0060	0.0002	0.0002	0.0000
352	347	0.0118	0.0064	0.0061	0.0002	0.0002	0.0000
358	353	0.0120	0.0065	0.0062	0.0002	0.0002	0.0000
364	359	0.0122	0.0066	0.0063	0.0002	0.0002	0.0000
370	365	0.0124	0.0067	0.0064	0.0002	0.0002	0.0000
376	371	0.0126	0.0068	0.0065	0.0002	0.0002	0.0000
382	377	0.0128	0.0069	0.0066	0.0002	0.0002	0.0000
388	383	0.0130	0.0070	0.0067	0.0002	0.0002	0.0000
394	389	0.0132	0.0071	0.0068	0.0002	0.0002	0.0000
400	395	0.0134	0.0072	0.0069	0.0002	0.0002	0.0000
406	401	0.0136	0.0073	0.0070	0.0002	0.0002	0.0000
412	407	0.0138	0.0074	0.0071	0.0002	0.0002	0.0000
418	413	0.0140	0.0075	0.0072	0.0002	0.0002	0.0000
424	419	0.0142	0.0076	0.0073	0.0002	0.0002	0.0000
430	425	0.0144	0.0077	0.0074	0.0002	0.0002	0.0000
436	431	0.0146	0.0078	0.0075	0.0002	0.0002	0.0000
442	437	0.0148	0.0079	0.0076	0.0002	0.0002	0.0000
448	443	0.0150	0.0080	0.0077	0.0002	0.0002	0.0000
454	449	0.0152	0.0081	0.0078	0.0002	0.0002	0.0000
460	455	0.0154	0.0082	0.0079	0.0002	0.0002	0.0000
466	461	0.0156	0.0083	0.0080	0.0002	0.0002	0.0000
472	467	0.0158	0.0084	0.0081	0.0002	0.0002	0.0000
478	473	0.0160	0.0085	0.0082	0.0002	0.0002	0.0000
484	479	0.0162	0.0086	0.0083	0.0002	0.0002	0.0000
490	485	0.0164	0.0087	0.0084	0.0002	0.0002	0.0000
496	491	0.0166	0.0088	0.0085	0.0002	0.0002	0.0000
502	497	0.0168	0.0089	0.0086	0.0002	0.0002	0.0000
508	503	0.0170	0.0090	0.0087	0.0002	0.0002	0.0000
514	509	0.0172	0.0091	0.0088	0.0002	0.0002	0.0000
520	515	0.0174	0.0092	0.0089	0.0002	0.0002	0.0000
526	521	0.0176	0.0093	0.0090	0.0002	0.0002	0.0000
532	527	0.0178	0.0094	0.0091	0.0002	0.0002	0.0000
538	533	0.0180	0.0095	0.0092	0.0002	0.0002	0.0000
544	539	0.0182	0.0096	0.0093	0.0002	0.0002	0.0000
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556	551	0.0186	0.0098	0.0095	0.0002	0.0002	0.0000
562	557	0.0188	0.0099	0.0096	0.0002	0.0002	0.0000
568	563	0.0190	0.0100	0.0097	0.0002	0.0002	0.0000
574	569	0.0192	0.0101	0.0098	0.0002	0.0002	0.0000
580	575	0.0194	0.0102	0.0099	0.0002	0.0002	0.0000
586	581	0.0196	0.0103	0.0100	0.0002	0.0002	0.0000
592	587	0.0198	0.0104	0.0101	0.0002	0.0002	0.0000
598	593	0.0200	0.0105	0.0102	0.0002	0.0002	0.0000
604	599	0.0202	0.0106	0.0103	0.0002	0.0002	0.0000
610	605	0.0204	0.0107	0.0104	0.0002	0.0002	0.0000
616	611	0.0206	0.0108	0.0105	0.0002	0.0002	0.0000
622	617	0.0208	0.0109	0.0106	0.0002	0.0002	0.0000
628	623	0.0210	0.0110	0.0107	0.0002	0.0002	0.0000
634	629	0.0212	0.0111	0.0108	0.0002	0.0002	0.0000
640	635	0.0214	0.0112	0.0109	0.0002	0.0002	0.0000
646	641	0.0216	0.0113	0.0110	0.0002	0.0002	0.0000
652	647	0.0218	0.0114	0.0111	0.0002	0.0002	0.0000
658	653	0.0220	0.0115	0.0112	0.0002	0.0002	0.0000
664	659	0.0222	0.0116	0.0113	0.0002	0.0002	0.0000
670	665	0.0224	0.0117	0.0114	0.0002	0.0002	0.0000
676	671	0.0226	0.0118	0.0115	0.0002	0.0002	0.0000
682	677	0.0228	0.0119	0.0116	0.0002	0.0002	0.0000
688	683	0.0230	0.0120	0.0117	0.0002	0.0002	0.0000
694	689	0.0232	0.0121	0.0118	0.0002	0.0002	0.0000
700	695	0.0234	0.0122	0.0119	0.0002	0.0002	0.0000
706	701	0.0236	0.0123	0.0120	0.0002	0.0002	0.0000
712	707	0.0238	0.0124	0.0121	0.0002	0.0002	0.0000
718	713	0.0240	0.0125	0.0122	0.0002	0.0002	0.0000
724	719	0.0242	0.0126	0.0123	0.0002	0.0002	0.0000
730	725	0.0244	0.0127	0.0124	0.0002	0.0002	0.0000
736	731	0.0246	0.0128	0.0125	0.0002	0.0002	0.0000
742	737	0.0248	0.0129	0.0126	0.0002	0.0002	0.0000
748	743	0.0250	0.0130	0.0127	0.0002	0.0002	0.0000
754	749	0.0252	0.0131	0.0128	0.0002	0.0002	0.0000
760	755	0.0254	0.0132	0.0129	0.0002	0.0002	0.0000
766	761	0.0256	0.0133	0.0130	0.0002	0.0002	0.0000
772	767	0.0258	0.0134	0.0131	0.0002	0.0002	0.0000
778	773	0.0260	0.0135	0.0132	0.0002	0.0002	0.0000
784	779	0.0262	0.0136	0.0133	0.0002	0.0002	0.0000
790	785	0.0264	0.0137	0.0134	0.0002	0.0002	0.0000
796	791	0.0266	0.0138	0.0135	0.0002	0.0002	0.0000
802	797	0.0268	0.0139	0.0136	0.0002	0.0002	0.0000
808	803	0.0270	0.0140	0.0137	0.0002	0.0002	0.0000
814	809	0.0272	0.0141	0.0138	0.0002	0.0002	0.0000
820	815	0.0274	0.0142	0.0139	0.0002	0.0002	0.0000
826	821	0.0276	0.0143	0.0140	0.0002	0.0002	0.0000
832	827	0.0278	0.0144	0.0141	0.0002	0.0002	0.0000
838	833	0.0280	0.0145	0.0142	0.0002	0.0002	0.0000
844	839	0.0282	0.0146	0.0143	0.0002	0.0002	0.0000
850	845	0.0284	0.0147	0.0144	0.0002	0.0002	0.0000
856	851	0.0286	0.0148	0.0145	0.0002	0.0002	0.0000
862	857	0.0288	0.0149	0.0146	0.0002	0.0002	0.0000
868	863	0.0290	0.0150	0.0147	0.0002	0.0002	0.0000
874	869	0.0292	0.0151	0.0148	0.0002	0.0002	0.0000
880	875	0.0294	0.0152	0.0149	0.0002	0.0002	0.0000
886	881	0.0296	0.0153	0.0150	0.0002	0.0002	0.0000
892							

Hydraulic oil injection method



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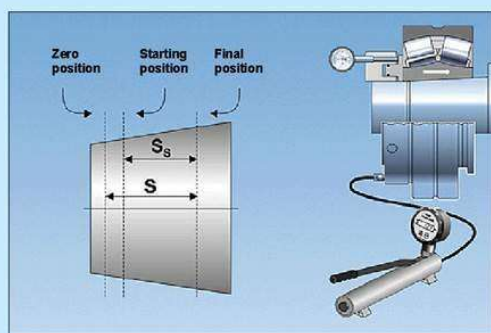
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Hydraulic mounting

The SRB drive-up method

The method...



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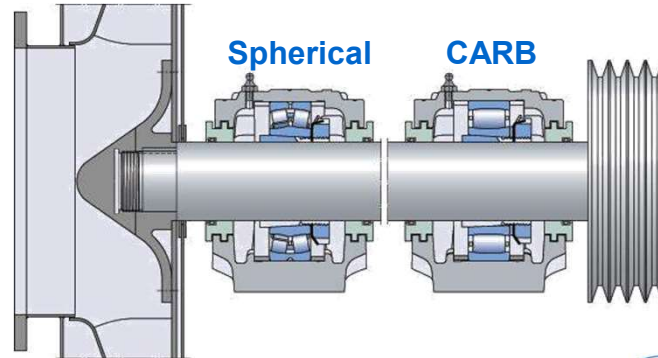
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SKF solutions for fans & blowers

- SKF Explorer spherical roller bearings
- Self-aligning ball bearings
- CARB toroidal roller bearings
- Adapter Sleeves
- SNL & SONL housings and seals



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Split housing installation

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Split housing installation

- Check surface flatness with precision straightedge
- Shim base with full shim stock until 0.050 mm (0.002") feeler gauge doesn't pass under housing
- Adjust housing base to prevent seal binding
- Torque grade eight bolts to recommended values if maximum housing strength is required

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Shimming the base

Under-coring allows uniform cooling when cast

Extended flats for dowel pin support



Base support in all critical areas

Flat, square foot for mounting and alignment

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Grease lubrication

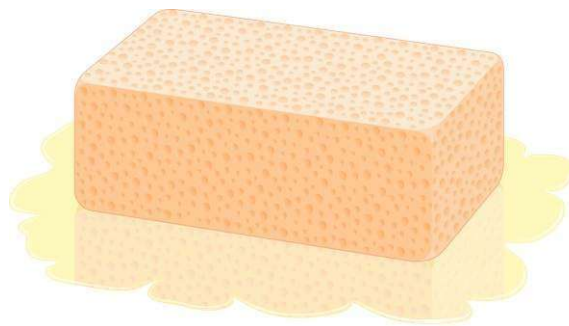
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What is grease?

- A grease is 80-90% oil held in a thickener (10-20%)



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What is grease not?

Grease is not:

- A thick oil
- A solid oil (wax)



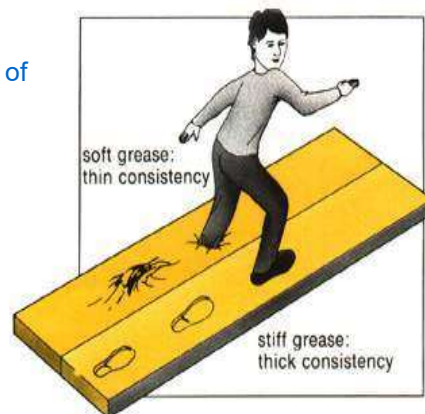
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Grease consistency...

- Consistency is the degree of stiffness that the grease possesses
- Consistency is specified in terms of a NLGI rating
- Consistency always refers to grease; Viscosity refers to oil



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Grease consistency

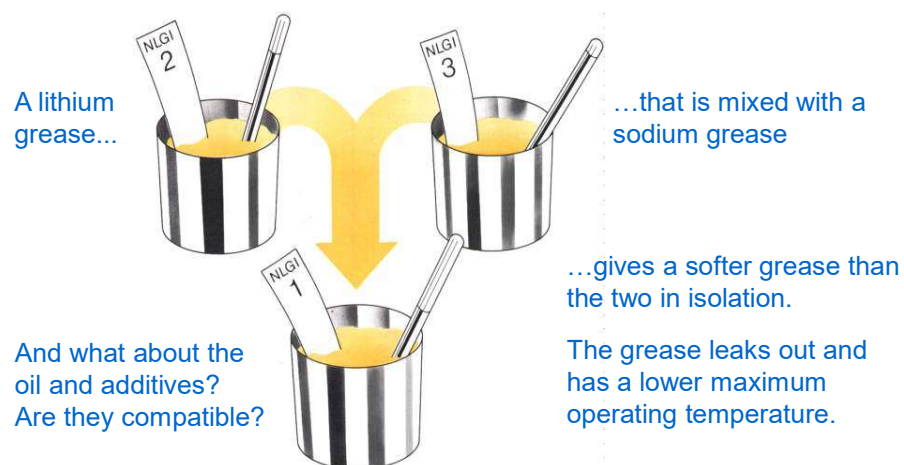
NLGI Number	Penetration [$\times 0.1$ mm]	Appearance	Use
000	445 - 475	Very Fluid	gear boxes etc.
00	400 - 430	Fluid	
0	355 - 385	Semi-Fluid	
1	310 - 340	Very Soft	bearings
2	265 - 295	Soft	
3	220 - 250	Medium Hard	
4	175 - 205	Hard	sealing
5	130 - 160	Very Hard	

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Incompatibility of greases



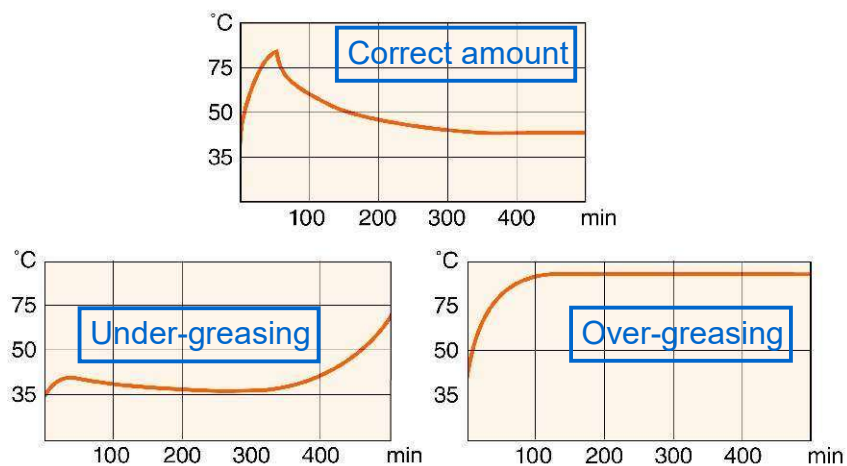
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Grease run-in



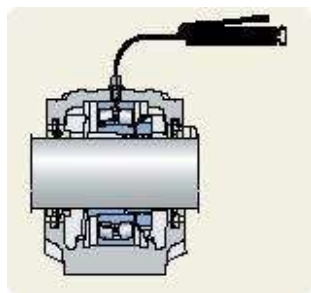
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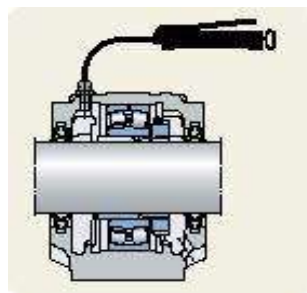
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Grease re-lubrication quantity...

$$G_p = 0.002 \cdot D \cdot B$$



$$G_p = 0.005 \cdot D \cdot B$$



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Grease re-lubrication quantity

- By adding small quantities of fresh grease at regular intervals, the used grease in the bearing arrangement will be partially replaced
- The re-grease quantity is based on:
 - **D**: bearing outside diameter, mm
 - **B**: bearing width, mm

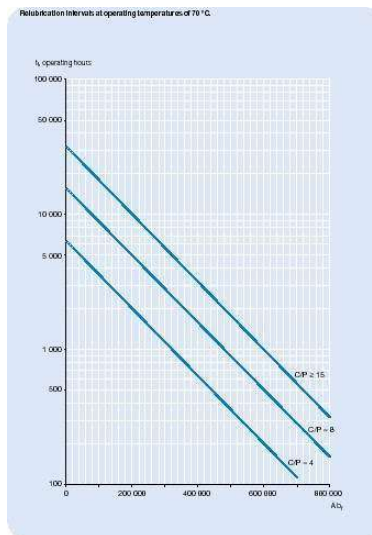
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Grease re-lubrication intervals...

Catalog 5000,
Page 238, Diagram 4



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Grease re-lubrication intervals

To calculate $A * b_f$:

- $A = n * d_m$
- n : rotational speed, r/min
- d_m : bearing mean diameter = $0.5(d+D)$, mm
- b_f : bearing factor depending on bearing type and load conditions

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SKF Dialset

Plug in your information:

- SKF Dialset gives you the answers

<http://www.mapro.skf.com/>

The screenshot shows the SKF Dialset software interface with the following sections:

- Bearing Basics:**
 - Dimensions: d = 100 mm, D = 190 mm, B = 42 mm
 - Type: ☒ DGBB/ACBB/SABB, ☒ SRB/CRB/CARB, ☐ TRB/TBB
- Operating Conditions:**
 - n-Speed: 2000 rpm, Op hrs/day: 24 hrs
 - Bearing Operating Temperature: ☒ Normal: 145 to 172 °F, ☐ Low: 117 to 145 °F, ☐ High: 172 to 199 °F, ☐ Very High: 199 to 225 °F
- Results:**
 - Interval: 1300 hrs, Relub - Corrected Interval: 200 hrs
 - Op - Grease Quantity: 0.5 oz, Feed rate: 0.06 oz/day
- Dial Setting SYSTEM 24®:** Shows a dial with the number 2.

At the bottom, there are buttons for 'Quit', 'LAGD 125', 'LAGD 60', and the SKF logo.

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SKF greases

SKF Maintenance and Lubrication Products MP3000E

Basic bearing grease selection

Generally use	LGMT 2	All purpose
Unless:		
Expected bearing temperature continuously > 100 °C/212 °F	LGHP 2	High temperature
Low ambient -50 °C/-58 °F, expected bearing temperature < 50 °C/122 °F	LGLT 2	Low temperature
Shock loads, heavy loads, vibrations	LGEP 2	High load
Food processing industry	LGFP 2	Food processing
'Green' biodegradable, demands for low toxicity	LGGB 2	Green biodegradable

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SKF System 24 automatic lubricator



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Common fan problems

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Objectives

At the end of this section you will have a better understanding of:

- Diagnosing common fan problems
- Practical Solutions

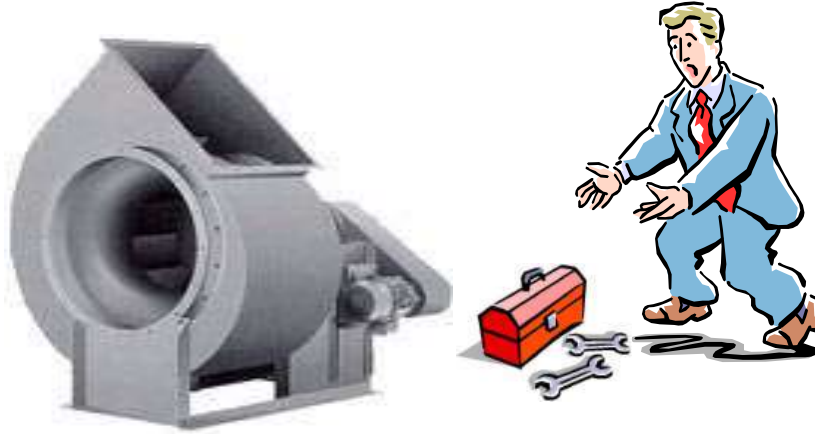
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What are your common fan problems?



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Common fan problems

- Out of balance
- Resonance
- Critical speeds
- Misalignment
- Soft foot
- Vibration
- Minimum load smearing in bearings
- Bearing spalling damage
- Lubrication issues
- Contamination

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More common fan problems

- Misalignment
- Free bearing not moving axially
- Bearings with improper clearance
- Seal rubbing friction (misaligned, wrong seals)
- Looseness (worn housing bore)
- Bolt torque not correct
- Lack of rigidity in the machine base
- Bearing installation damage

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Fan system management program

- Vibration monitoring
- Lubrication analysis
- Advanced diagnostic analysis
- Bearing and housing installation
- Precision alignment
- Precision in-situ dynamic balancing
- Lubrication recommendations
- Seal application review
- Bearing system design upgrades

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Unbalance

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Out of balance



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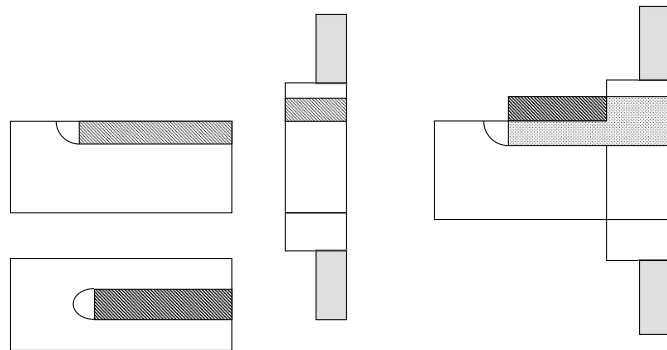
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Out of balance: assembly practices...

Key Length



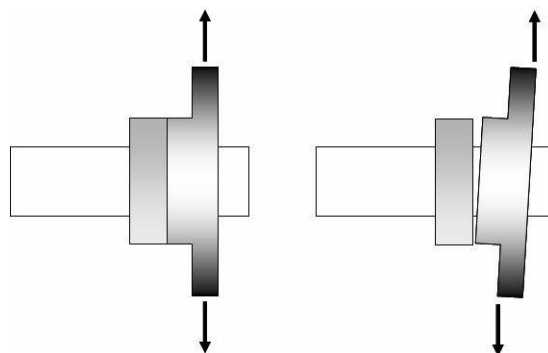
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Out of balance: assembly practices...

Cocked Rotor



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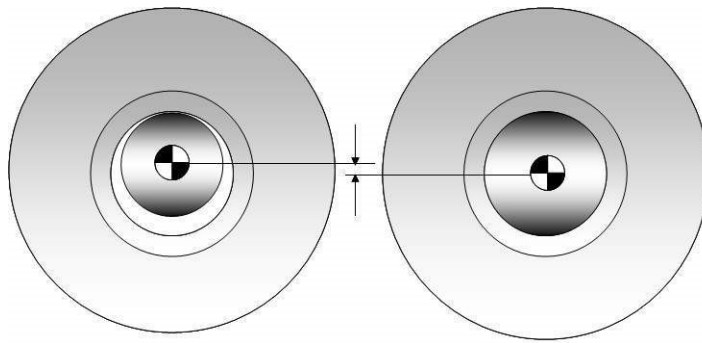
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Out of balance: assembly practices

Slip fits/change in centerline position

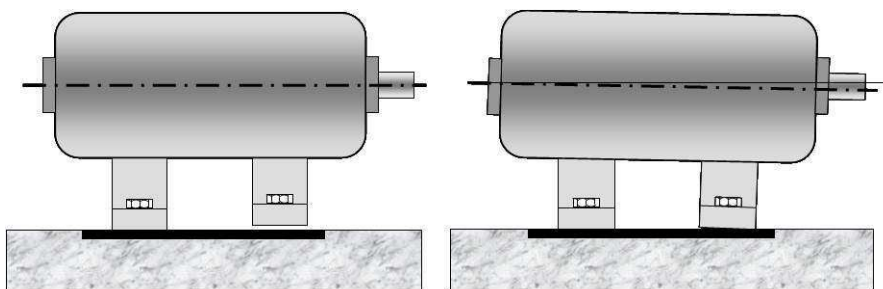


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Out of balance: soft foot



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Resonance

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Resonance

Exists in every machine

- All machines have natural frequencies
- Only a problem when a resonant frequency matches a forcing frequency produced by a machine or process
- Frequency separation of 30% or less can cause problems

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Resonance example



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Resonance example case

- History of high vibration since originally installed
- Localized vibration at one location and direction – horizontal inboard pump bearing
- Previous attempts at correcting the problem have been unsuccessful
 - New coupling
 - Pipe bracing

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Four components of resonance

1. Mass
 2. Stiffness
 3. Force
 4. Damping
- The combination of all four determines if a machine or structure will RESONATE and how much amplification will occur

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Mass and stiffness

- Combination determines the natural or resonant frequencies
- Increasing mass decreases a resonant frequency; decreasing mass raises it
- Increasing stiffness raises a resonant frequency; lowering stiffness lowers it

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Mass and stiffness changes

Check history to identify changes to a machine or structure

- Motor replacement with a smaller or larger motor – especially on vertical machines
- Changes to foundation and support – grouting, filling with cement, cutting or adding bracing
- Pipe support changes – adding, moving, removing

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Force

- If there is no force, a part will not resonate
 - A bell will not ring unless struck or excited
- Number one way to reduce or eliminate a resonance problem is to attack the force
- This is usually the last thing attempted

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Damping

- Damping controls the amplification of the vibration due to resonance and can be designed in a system to allow for speed changes, etc.
- Shock absorbers (dampers) on cars do not eliminate resonance, they control the amplification and provide rapid decay of the vibration

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Critical speeds

- Critical speeds are a special case of resonance
- Critical speeds are rotating speeds that excite a rotor resonance
- Rotor does not vibrate back-and-forth, it begins to “whip”
- Can be very damaging
- Should not be confused with structural resonances

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Misalignment

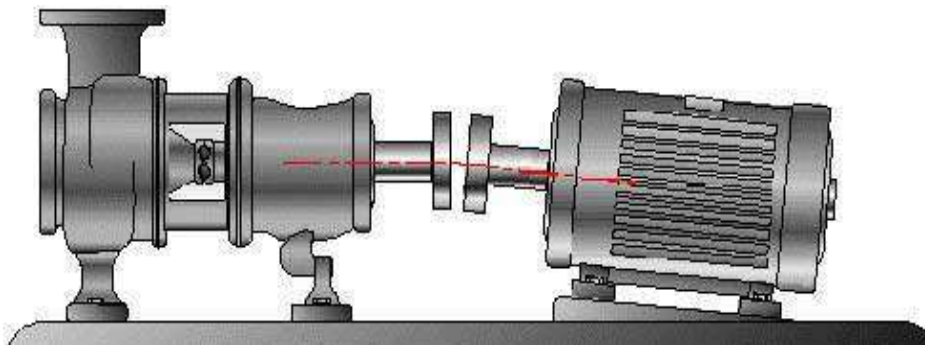
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Misalignment

- A proactive alignment program can give a fast and large payback in reduced downtime on the fan farm



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Why align?

Improve machine performance:

- Decrease power consumption
- Decrease noise and vibration
- Prevent premature bearing wear
- Extend the life of seals
- Lower coupling wear

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Precision alignment

- Detection
 - Amplitude
 - Frequency
 - Phase
- Machine History

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Misalignment

- Misalignment is characterized by different motion in one radial plane versus the other

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Detecting misalignment

- Highest amplitudes often point to the direction/plane of the misalignment
- Hand-feel of the amplitudes
- Further inspection
 - Base
 - Feet
 - Surrounding equipment/machines

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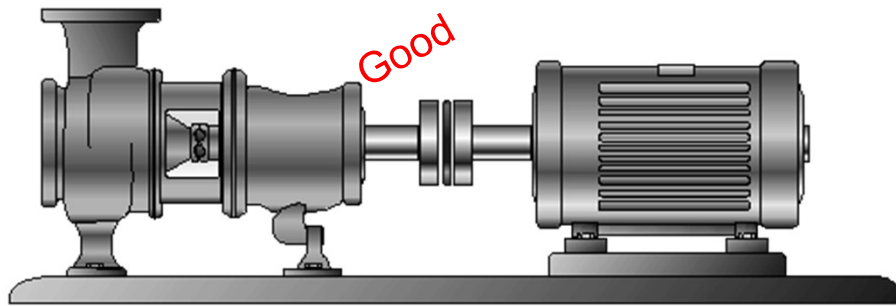
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Machinery alignment

- Ideally, shafts are collinear at operating temperature



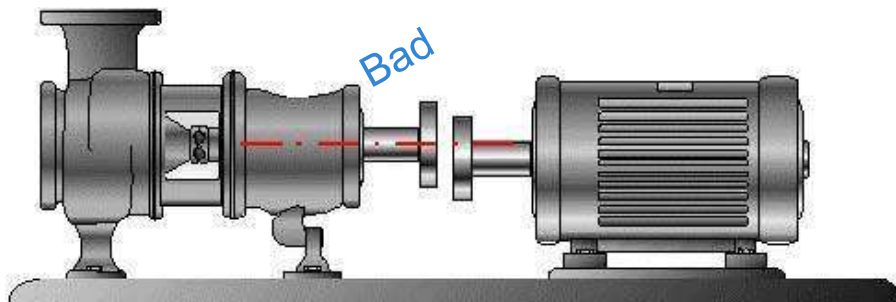
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Offset/parallel misalignment

- Shafts are offset, but parallel



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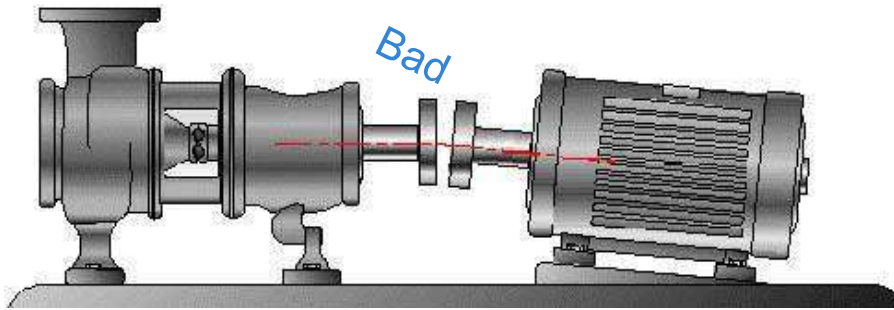
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Angular misalignment

- Zero offset at coupling



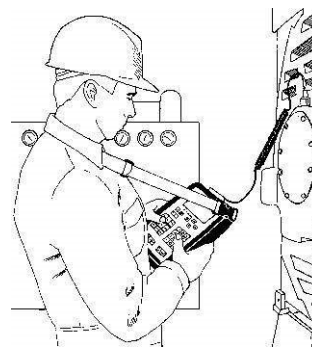
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Vibration spectra

- Characteristics of unbalance versus misalignment



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Running speed amplitude and multiples

- Progression from unbalance to misalignment
- Why machines fail to read the books
- Correlation between amplitudes and frequencies

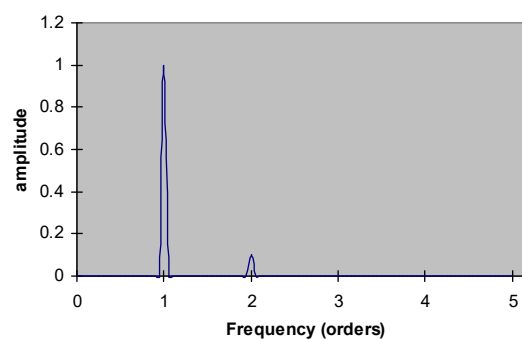
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Unbalance

- Higher than usual amplitude at running speed
- Small amplitudes at the harmonics
- Two-times running speed amplitude < 30% of running speed amplitude



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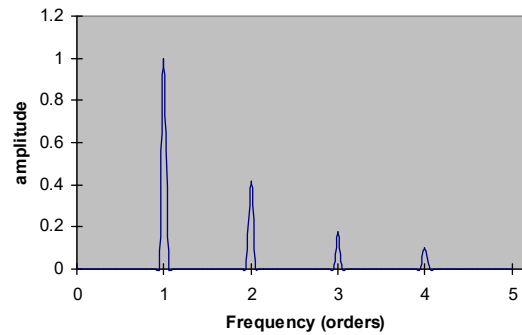
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Unbalance towards misalignment

- Amplitude at two-times running speed between 30% and 50% of running speed
- Possible harmonics out to approximately 5 times running speed
- Hard to determine if the machine is “mostly” out of balance or “mostly” misaligned



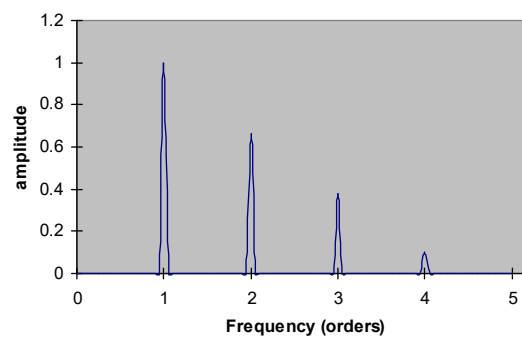
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More likely misalignment

- Amplitude at two-times running speed between 50% and 100% of running speed



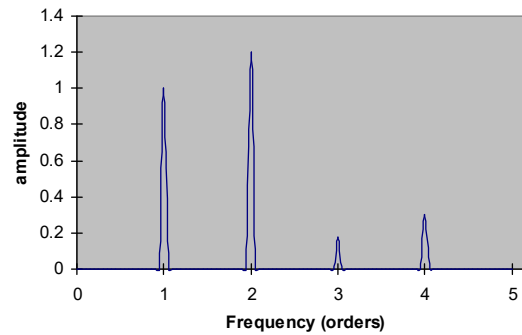
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Possible resonance

- If amplitude at two-times running speed exceeds one-times running speed, suspect resonance
- Structural dynamics of machine, coupling type, and type of misalignment will affect the various harmonic amplitudes



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Phase analysis

- Phase improves accuracy of diagnosis to 95%
- Phase reveals other machinery problems mistaken for unbalance or misalignment or that aggravate the condition

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Machine history...

- Has the machine always vibrated excessively or did the problem recently develop?
- What maintenance has been performed recently on the machine, or can any maintenance action be correlated to when there was a change in machine behavior?
- How does the machine behave when first started to the point where it reaches normal operating temperatures?

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Machine history

- Do the symptoms change with process or ambient temperature changes?
- What alignment method was used?
- What do the alignment records show for the as-found and as-left positions?
- What thermal growth values are used and how were they computed?
- What are the historical failure rates of bearings, seals, coupling components, or other wear items?

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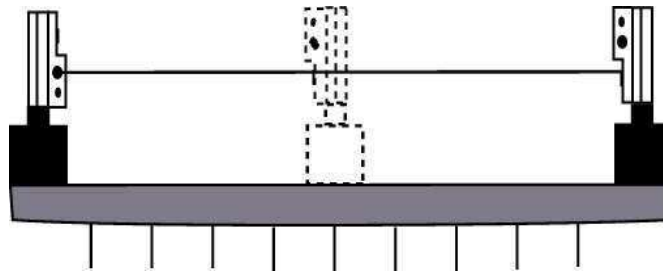


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Alignment tolerances...

Bases:

- The recommended tolerance is 0.025 mm (0.001") for all mounting pads, and the feet of the machines should be coplanar to within 0.025 mm (0.001")



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Alignment tolerances...

Deflection due to piping or conduit:

- The recommended precision tolerance is to limit shaft deflection to 0.05 mm (0.002") at the centerline of the coupling in any direction when piping or conduit is attached

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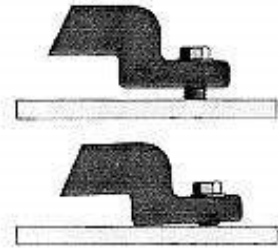
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Alignment tolerances...

Soft Foot:

- The gap under all feet should not exceed 0.025 mm (0.001")



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Alignment tolerances

Offset and Angle:

- Maximum offset at the coupling – 0.025 mm (0.001")
- Maximum offset at the non-drive end feet (either machine) – 0.05 mm (0.002")

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Shimming



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Conduit strain



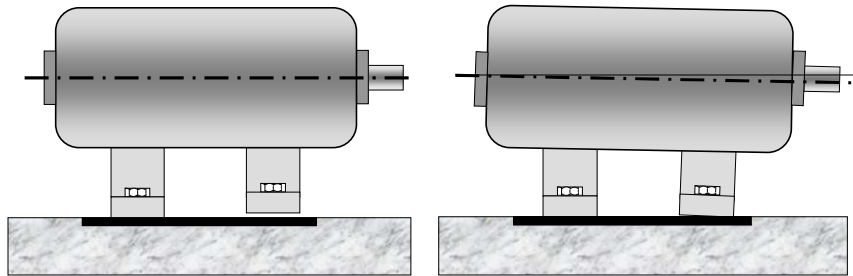
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Soft foot: frame distortion



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Fan bearing lubrication

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Grease lubrication for fan bearings

- Generally, for split pillow blocks with SRB's and SABB's, fill the bearing 100% full of grease
- CARB: fill bearing 1/2 full, housing 1/3 full, and labyrinth seal grooves 100%
- Recommended grease:
 - Lithium or Lithium complex thickener
 - NLGI Grade 2
 - Oil in grease typically ISO VG 150 to 220 (depending on bearing operating temperature)
 - Generally, rust and oxidation inhibitors are desirable
- Re-lubrication mainly depends on the contamination levels and purging action of the seals

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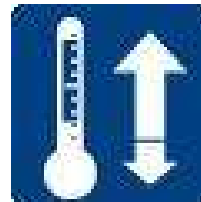


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Typical fan bearing grease LGHP2

High performance, wide temperature, long life, high speed polyurea bearing grease

- Electric motors
- Hot fans
- Water pumps



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Lubrication selection guidelines

Ambient gas fans:

- Mineral oil based grease up to 180° F bearing temperature
- Synthetic oil based grease up to 220° F bearing temperature
 - Mobilith SHC 220 or equivalent is very common for SRB's operating at 180° F to 220° F
- For above 220° F (Bearing Operating Temperature), circulating oil is recommended
- Hot gas fans (air/gas temp > 300° F) typically require circulating oil lubrication

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Fans: oil viscosity general guidelines

- Spherical roller bearings require minimum 21 cSt at bearing operating temperature
- Self-aligning ball bearings require minimum 13 cSt at bearing operating temperature
- For most ambient temperature air/gas fans, ISO VG 100 to 220 base oils are best

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Automatic grease re-lubrication



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Contamination...

Abrasive Wear:

- Formed by large quantities of very small contaminant particles in the lubricant
- Will subject the bearing to continuous polish wear



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Contamination

- This bearing had sand and rock dust throughout the bearing and in the lubricant
- There was extensive abrasive wear on all components
- The seals had failed



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Contamination prevention

- Keep the bearings clean while mounting
- Keep the lubricant clean
- Select the right seals
- Use “Taconite” (extreme service seals) where necessary
- Install seals correctly
- Clean off the grease nipples and grease gun valve
- Don't contaminate when you re-lubricate
- Grease and purge where necessary
- Clean the oil sump
- Filter the oil if a circulating oil system is used

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Bearing installation damage

- Spalling caused by impact damage during mounting



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Fan problem solving

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Rigidity in the machine base

Structure maintenance:

- All structures used to support fans should be checked regularly
- Check vibration isolators, bolts, and foundations for corrosion and erosion
- Check for corrosion, cracks, and other signs of stress
- Check structural integrity of concrete foundations

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Foundations and support structures...

Fan foundations should be:

- Flat
- Level
- Rigid
- Made of concrete with a mass 3 to 5 times that of the fan assembly

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Foundations and support structures...

Anchor bolts:

- Should be “L” or “T” shaped long enough for nuts, washers, shims, and threads for draw-down
- Bolt should be in a sleeve or pipe with diameter larger than the bolt for adjustment



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Foundations and support structures

Fans mounted within a structure:

- Should be placed as close as possible to a rigid member, such as a wall or column
- The structure must be designed for rotating equipment, not just static design
- Supports should be cross-braced to prevent side sway
- Structural resonance frequencies at least +/- 20% of fan speed
- Vibration isolators should be used where applicable

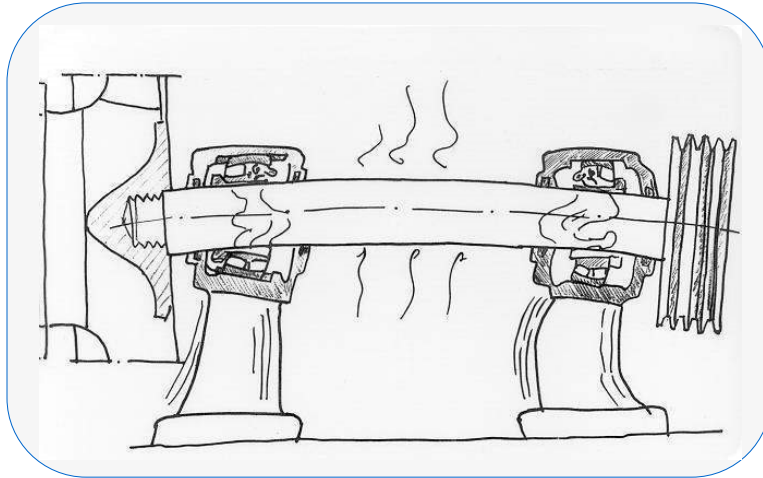
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Free bearing not moving axially



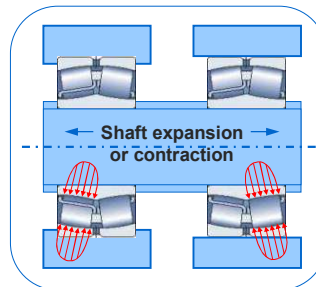
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Worst case scenario: the non-locating bearing cannot move

- The bearings become axially preloaded
- As the shaft expands or contracts, very high axial forces are generated between the bearings



Operating temperature increases and there is a high risk for bearing failure

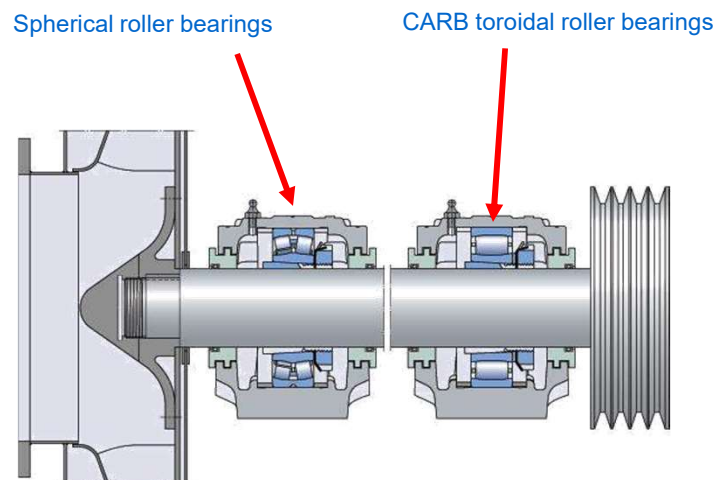
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SKF self-aligning system



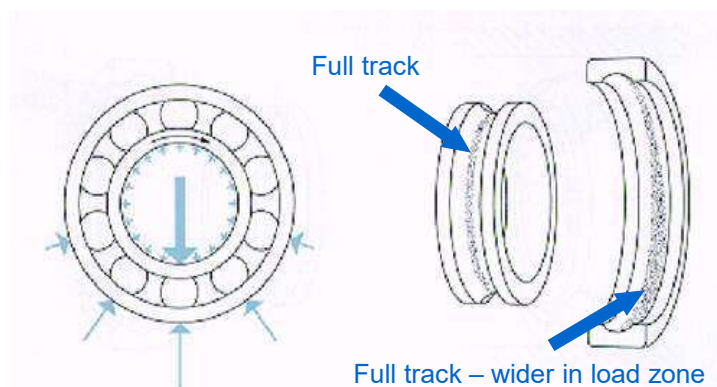
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Bearings with too little internal clearance

Rotating inner ring: fits too tight



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Seal malfunction

Observed Problem:

- Housing running hot
- Oil or grease leakage
- Noise
- Rough running

Potential Cause:

- Wrong type of seals selected
- Seals rubbing
- Seals misaligned
- Seals worn

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Looseness: bearing to housing fretting corrosion

- Fretting corrosion between bearing outer ring and housing
- This is a common problem since fretting corrosion often will be formed after some time in service



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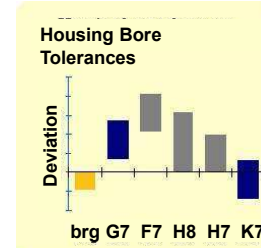
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Looseness: bearing to housing – cure

- Shaft tolerances are generally h9/IT5
- Housing fits for clearance fit - G7
- For intermediate/tight fit – K7 (Self-Aligning System)
- Benefits:
 - Elimination of housing wear and fretting corrosion
 - Improved reliability



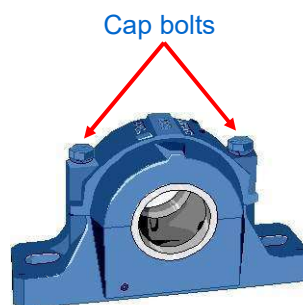
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Bolt torque incorrect

- Pillow block housing bolts must be torqued
- Check catalog for correct torque limits
- Use a torque wrench



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Module closeout

2025-05-29 © SKF® Slide 182 WE 202
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Name: _____ <Optional to Include Your Name>

WE 202 - Bearings in Rotating Machinery Applications
XX/XX/2007

Module Number & Title: 4. Industrial Fans: Improving System Reliability

My **most important question on this module** is:

.....
.....
.....

I **did not understand** the following:

.....
.....
.....

I **do not agree with** the following:

.....
.....
.....

My **top three learnings from this module (referenced to your expectations)** are:

1.
.....
2.
.....
3.
.....

Is this learning related to
your own expectations?

Yes No

Yes No

Yes No

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